A. Balanced Rating Changes

time limit per test 1 second
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 1000 | Tags: implementation, math | https://codeforces.com/problemset/problem/1237/A

Another Codeforces Round has just finished! It has gathered n participants, and according to the results, the expected rating change of participant i is a_i. These rating changes are *perfectly balanced* — their sum is equal to 0.

Unfortunately, due to minor technical glitches, the round is declared *semi-rated*. It means that all rating changes must be divided by two.

There are two conditions though:

- For each participant i, their modified rating change b_i must be integer, and as close to $(a_i)/2$ as possible. It means that either $b_i = \lfloor (a_i)/2 \rfloor$ or $b_i = \lceil (a_i)/2 \rceil$. In particular, if a_i is even, $b_i = (a_i)/2$. Here $\lfloor x \rfloor$ denotes rounding down to the largest integer not greater than x, and $\lceil x \rceil$ denotes rounding up to the smallest integer not smaller than x.
- The modified rating changes must be perfectly balanced — their sum must be equal to 0.

Can you help with that?

Input

The first line contains a single integer n ($2 \le n \le 13\,845$), denoting the number of participants.

Each of the next n lines contains a single integer a_i ($-336 \le a_i \le 1164$), denoting the rating change of the i-th participant.

The sum of all a_i is equal to 0.

Output

Output n integers b_i, each denoting the modified rating change of the i-th participant in order of input.

For any i, it must be true that either $b_i = \lfloor (a_i)/2 \rfloor$ or $b_i = \lceil (a_i)/2 \rceil$. The sum of all b_i must be equal to 0.

If there are multiple solutions, print any. We can show that a solution exists for any valid input.

Examples

Input
3 10 -5 -5

Output

```
5
-2
-3
```

Input

```
7
-7
-29
0
3
24
-29
38
```

Output

```
-3
-15
0
2
12
-15
19
```

Note

In the first example, $b_1 = 5$, $b_2 = -3$ and $b_3 = -2$ is another correct solution.

In the second example there are 6 possible solutions, one of them is shown in the example output.